2026 | AIMAHEAD AIHEC TCU student AI/ML Readiness Program

Unraveling the Patterns

PREDICTING FREQUENT MENTAL DISTRESS ACROSS SURVEY YEARS AND ASSESSING MODEL RELIABILITY FOR AMERICAN INDIAN AND ALASKA NATIVE ADULTS



presented by

Kristine Allison Johnson



Mental Health as a Public-Health Priority

In 2022, 15.9% of U.S. adults reported frequent mental distress, defined as 14 or more mentally unhealthy days during the preceding 30 days.
Source: CDC, 2024

Frequent mental distress is a population-health indicator and not a clinical diagnosis.

Public-health surveillance can help identify patterns in community health and highlight where additional support and resources may be needed.

Mental Health in American Indian and Alaska Native Communities

Measure, 2024	AI/AN adults*	U.S. adults overall
Any mental illness in the past year	25.90%	23.40%
Received mental-health treatment in the past year	19.10%	22.90%

AI/AN adults were **approximately 17% less likely than U.S. adults overall to have received mental-health treatment.**

Source: HHS Office of Minority Health, using 2024 SAMHSA and CDC survey estimates.

What Does Existing Research Tell Us?

Singh et al. (2024) examined social determinants and health disparities using multiple national data sources.

Their 2018–2020 BRFSS analysis found higher frequent mental distress among AI/AN adults than among U.S. adults overall. pmc.ncbi.nlm.nih

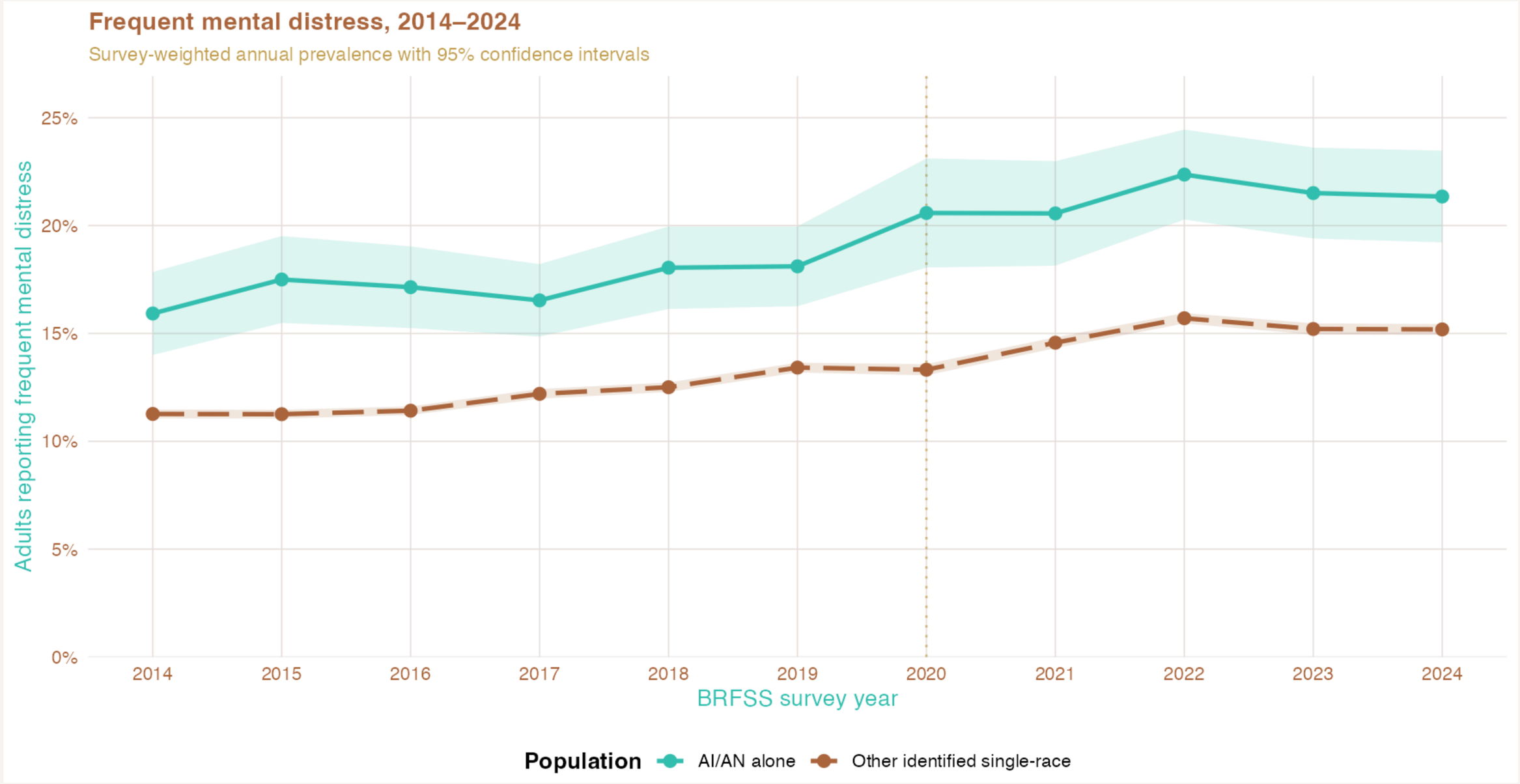
19.1% AI/AN adults

13.3% total US adults

Frequent Mental Distress, 2018–2020

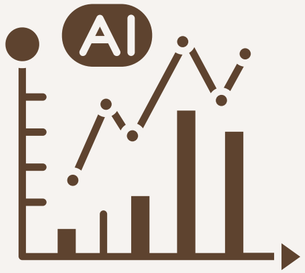
Difference: 5.8 percentage points

Source: Singh et al. (2024), International Journal of Maternal and Child Health and AIDS.



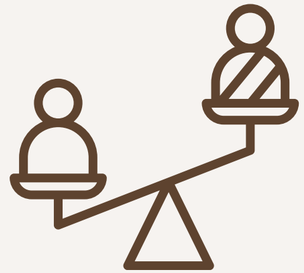
Source: Analysis of BRFSS, 2014–2024 for this project

Background & Motivation



Predictive Analytics

Machine learning can help identify patterns in health data that can be useful for understanding community needs, planning services, and directing resources.



Existing Mental-Health Disparities

American Indian and Alaska Native (AI/AN) adults experience differences in mental-health outcomes and access to care, highlighting the need for research that examines population-specific health patterns (HHS OMH, 2026).

Social Determinants and Health Factors Informing the Study

A conceptual framework for organizing domains and selecting available BRFSS variables



Socioeconomic Circumstances

Income
Education
Employment
Housing



Physical health and chronic conditions

General health
Physically unhealthy days
Chronic illness
Functional limitations



Historical, structural, and community factors

Historical trauma
Discrimination
Cultural continuity
Community support and belonging

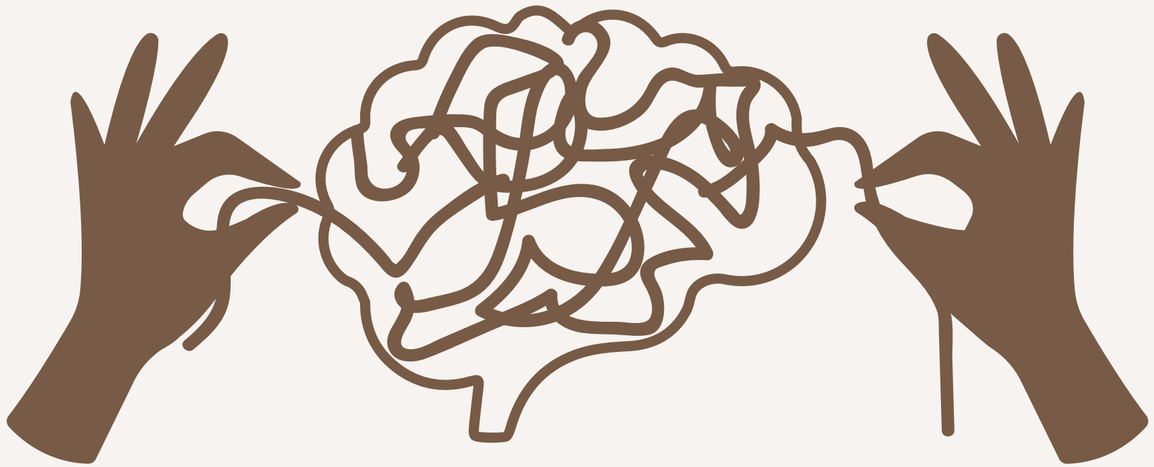


Access to healthcare and services

Access to Healthcare Providers
Affordability
Routine care

Conceptual framework informed by Healthy People 2030 and Singh et al. (2024); physical-health measures included as complementary study domains.

Background & Motivation



Predictive models learn from data shaped by existing differences in health, access to care, and use of services. Previous research shows that these patterns can influence which people a model identifies as needing support (Obermeyer et al., 2019).

When models are used to inform services or resource planning, relying only on overall predictive performance may overlook differences in how well they recognize need across populations. This creates a risk of reinforcing existing gaps rather than addressing them.

For this reason, I evaluated how the same nationally trained model performs across study populations, with a focus on American Indian and Alaska Native adults.

Dataset Overview

Data Source

Behavioral Risk Factor Surveillance System (BRFSS)

A CDC-supported, state-based annual survey collecting self-reported information on adult health, healthcare behaviors, chronic conditions, and preventive care.

Study Period

2023 – 2025

AI/AN respondents (American Indian/Alaska Native)

Respondents by Year

Year	Respondents
2023 (training)	8,951
2024 (comparison)	9,156
2025 (comparison)	6,973
Total	**25,080**

Unsupervised Learning Data Preparation Pipeline

449

original variables

Raw questionnaire variables, fields and identifiers

25

selected predictors

Health and social variables across 7 domains

Preprocessing

One-hot encoding, scaling, and domain balancing

89

encoded features

Numerical columns supplied to PCA and k means

Selected Predictors after Preprocessing

2023 – 2025

all AI/AN respondents (American Indian/Alaska Native)

physical health

GENHLTH - General Health
PHYSHLTH - Physically Unhealthy Days

chronic conditions

ASTHAM3 - Asthma
CVDINFR4 - Heart Attack
CVCRHD4 - Heart Disease
CVSTRK3 - Stroke
CHCCOPD3 - COPD
CHCKDNY2 - Kidney Disease
HAVARTH4 - Arthritis
DIABETE4 - Diabetes

health activity

EXERANY2 - Excercise during the past 30 days

functional health

DIFFWALK - Difficulty Walking
DIFFDRES - Difficulty Dressing
DIFFALON - Doing errands alone
DEAF - Serious hearing difficulty
BLIND - Serious vision difficulty

socioeconomic factors

EDUCA - Education
INCOME3 - Household Income
EMPLOY1 - Employment
RENTHOM1 - Home Tenure



Mental health variables were deliberately excluded from clustering, FMD was examined afterwards to determine how it differed between the identified profiles.

healthcare access

PERSDOC3 - Personal Healthcare Provider
MEDCOST1 - Inability to Afford Medical Care
CHECKUP1 - Time Since Routine Checkup

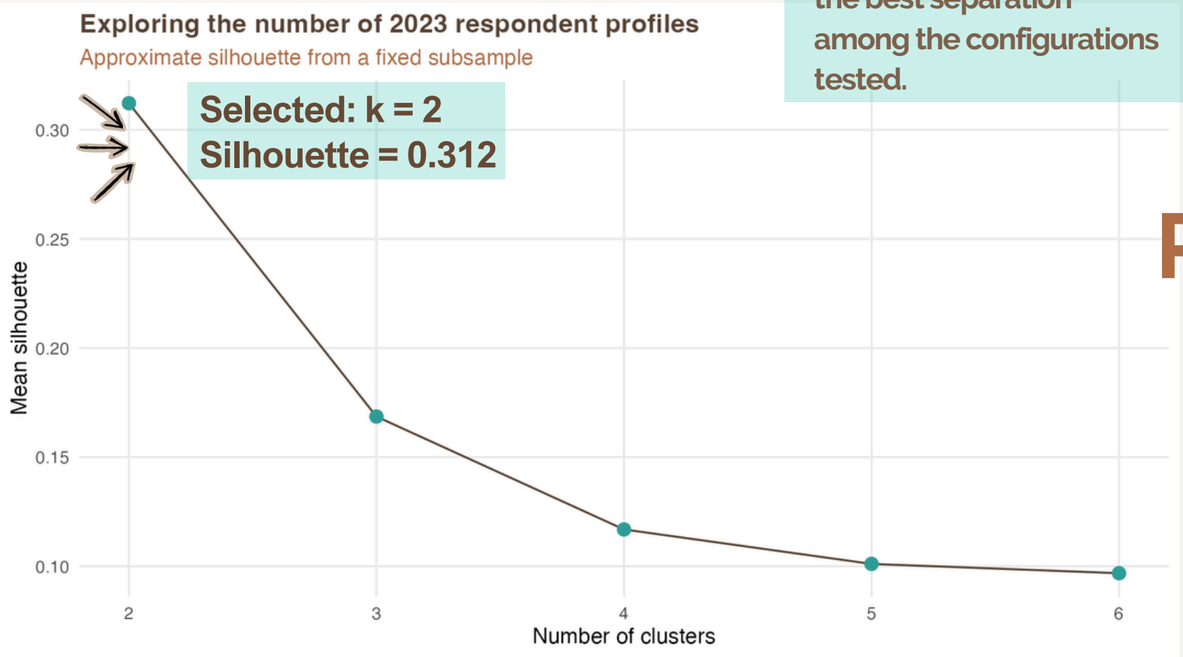
demographics

_AGEG5YR - Age Group
SEXVAR - Sex

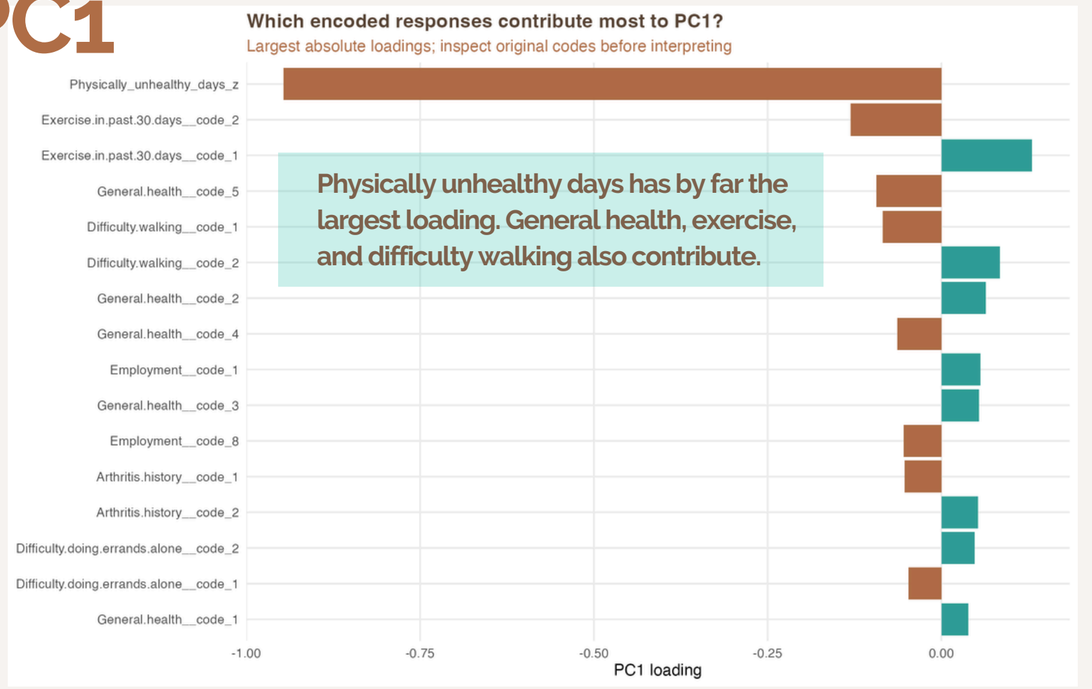
Principal Component Analysis (PCA) & K-Means Clustering

2023 BRFSS | AI/AN dataset

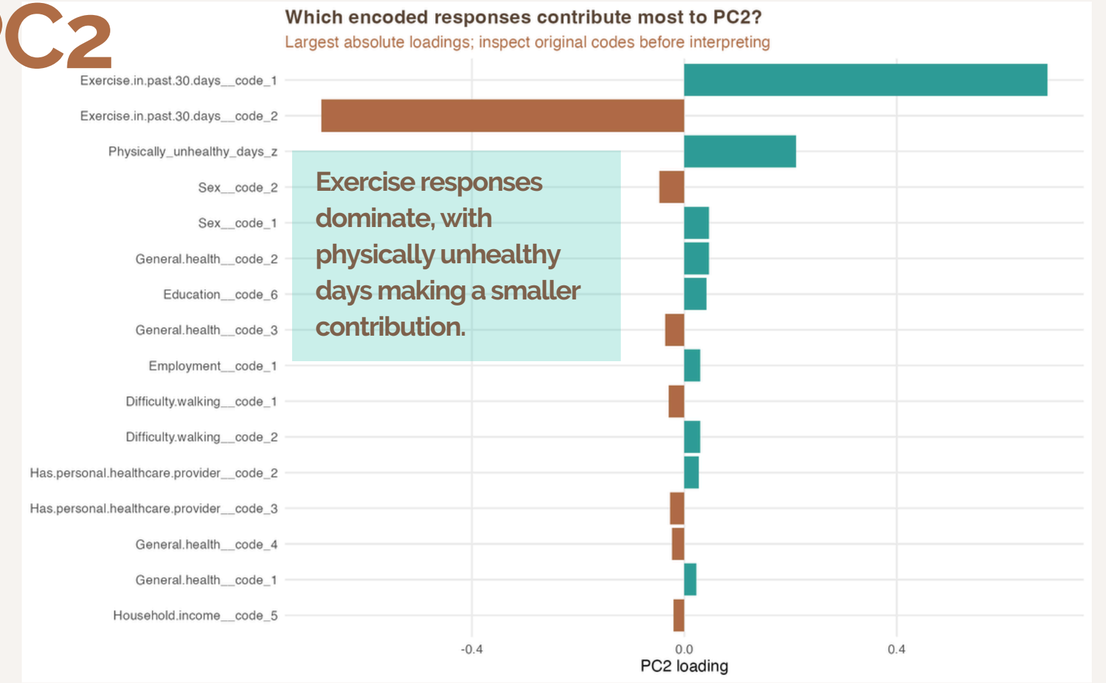
A two-cluster solution produced the highest approximate silhouette score (0.312), indicating the best separation among the configurations tested.

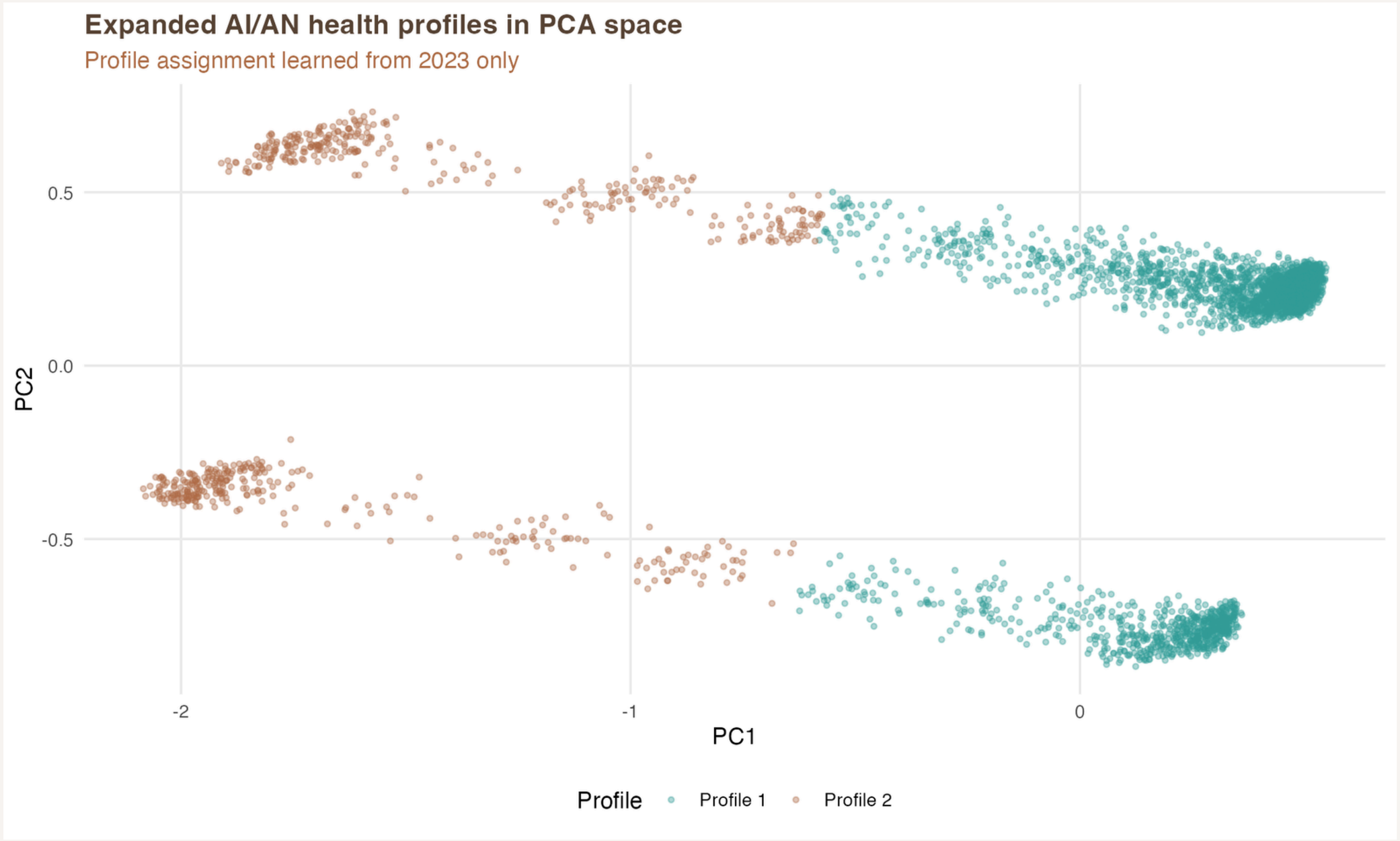


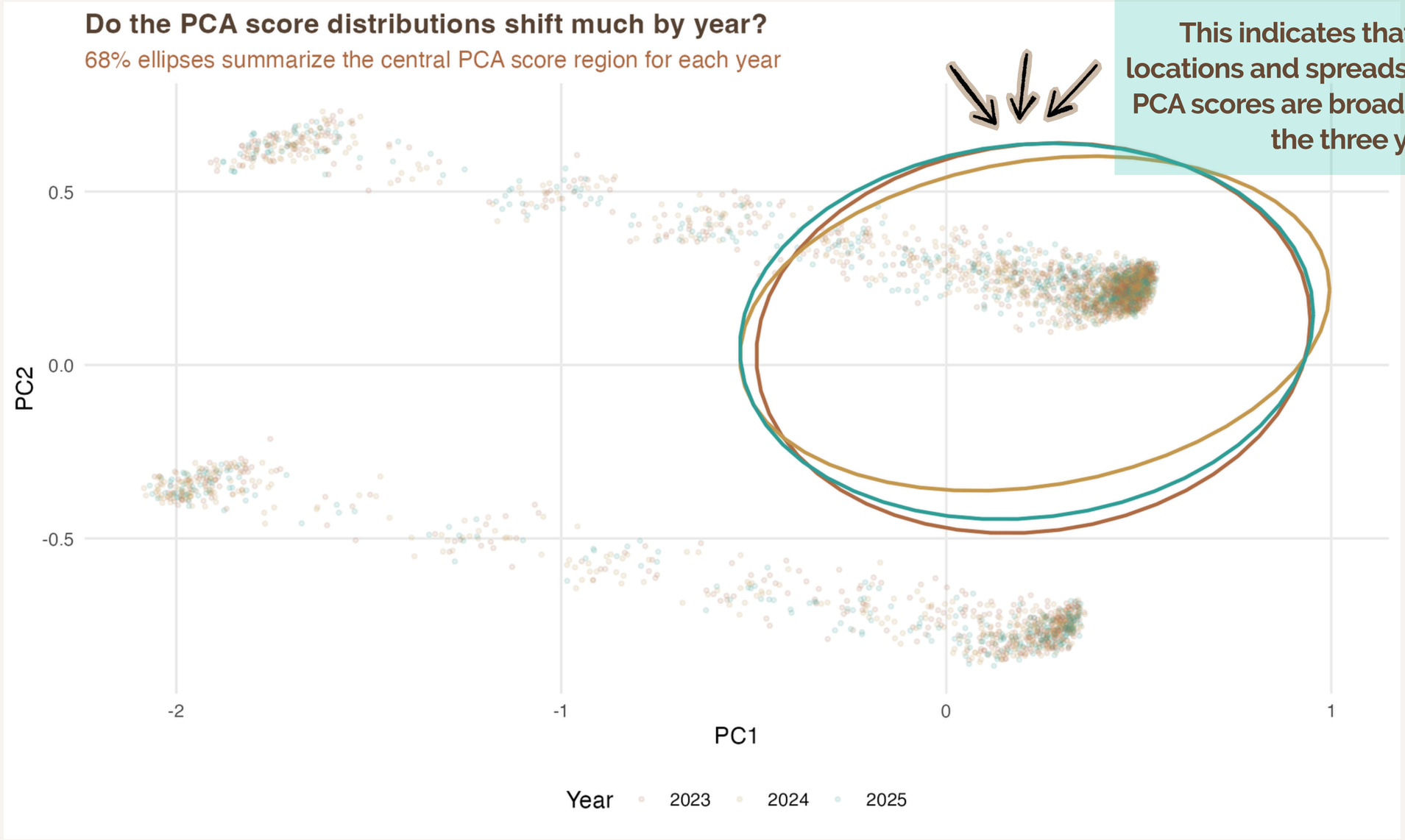
PC1



PC2




Expanded AI/AN health profiles in PCA space
Profile assignment learned from 2023 only
0.5
0.0
-0.5
PC2
-2
-1
0
PC1
Profile
Profile 1
Profile 2



This indicates that the central locations and spreads of respondents' PCA scores are broadly similar across the three years.

2023, 2024, and 2025 BRFSS | AI/AN dataset

Conclusions from PCA

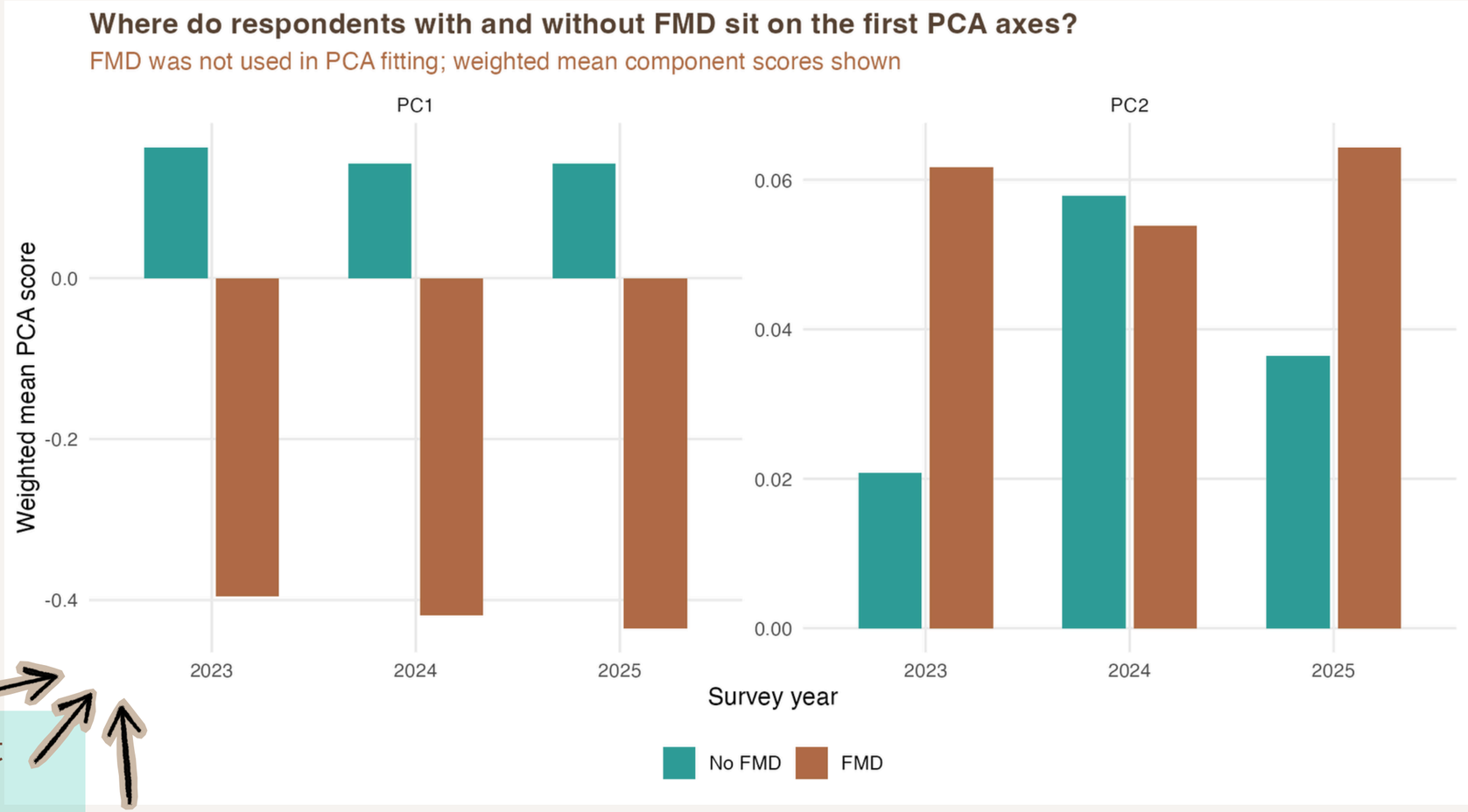
physical health

PC1: Physical-Health

Respondents with FMD have substantially lower (more negative) PC1 scores than those without FMD in every year.

GENHLTH - General Health
PHYSHLTH - Physically Unhealthy Days

FMD aligns with the main PCA pattern (PC1), indicating a potential link between physical health and mental distress.



Although FMD was not used in the model, it aligns with clear differences on PC1.

What This Tells Us

A principal component dominated without mental-health outcomes still distinguishes respondents with and without FMD, driven primarily by health-related PC1.

health activity

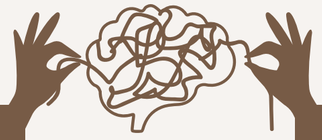
PC2: Primarily captures Exercise Behavior

FMD differences on PC2 are much smaller and less consistent across years.



Research Question

How well do machine-learning models trained on **historical BRFSS data** classify **self-reported frequent mental distress** in later survey years, and how does **predictive performance differ** between American Indian and Alaska Native (AI/AN) adults (p1) and other adult populations(p2)?



question 1

Can we predict frequent mental distress?

Can models trained on earlier BRFSS data accurately classify self-reported frequent mental distress among respondents in later survey years?

question 2

How well does the model work for AI/AN adults?

How does the same nationally trained model perform for AI/AN-adults(p1) compared with other self-identified single-race adults(p2)?

Dataset & Study Population



Data Source: Behavioral Risk Factor Surveillance System (BRFSS)

A CDC-supported, state-based annual survey collecting self-reported information about adult health, health behaviors, chronic conditions, and preventive care.



Study Period: 2014–2025 (12 annual surveys)

Variables used:

- physical-health days
- general health
- age
- gender
- education
- income



Historical Dataset (2014–2024):

4,790,742 processed respondent records



2025 Evaluation Dataset:

356,158 respondent records

6,793 AI/AN | 323,989 other



Frequent Mental Distress

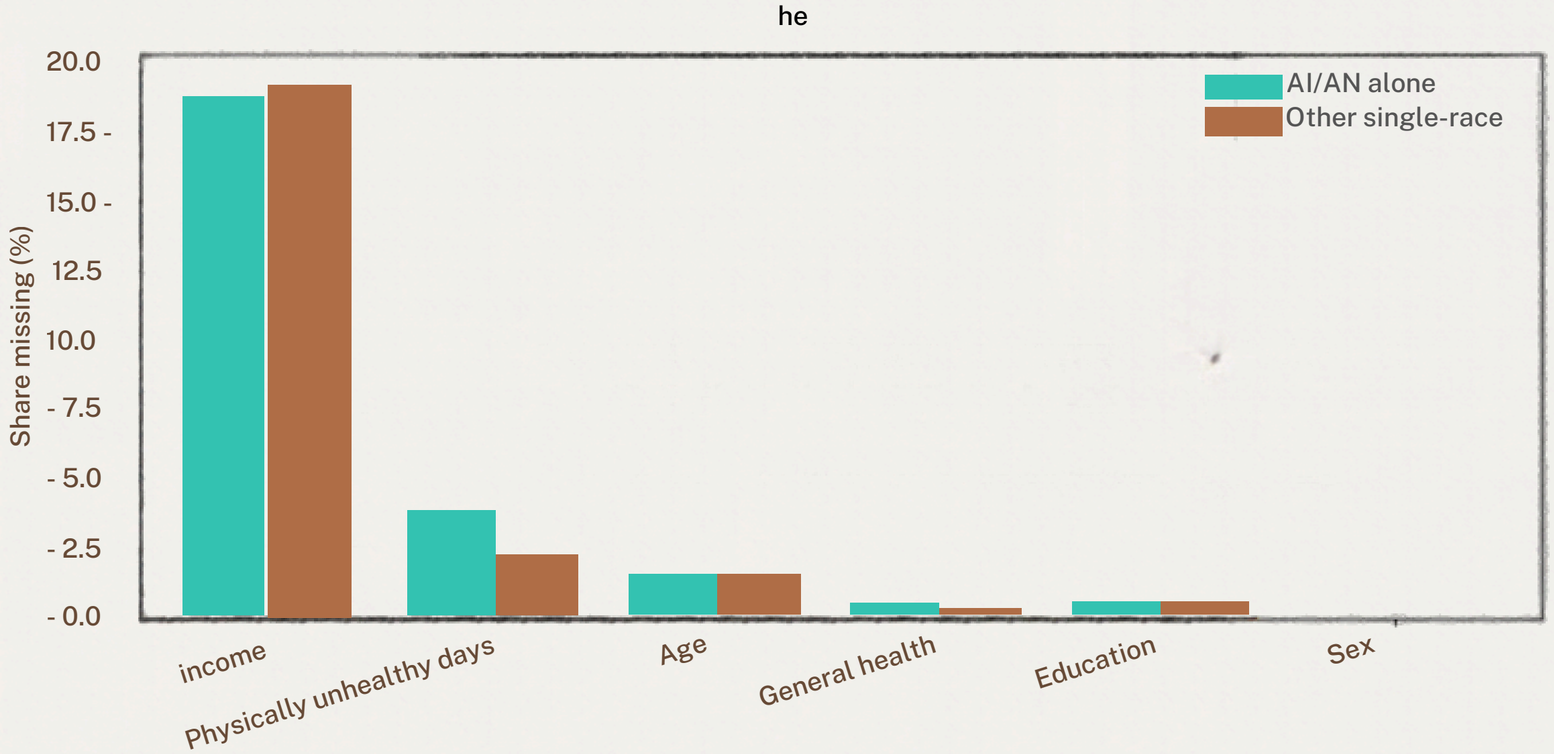
Frequent Mental Distress (FMD), defined as 14 or more self-reported mentally unhealthy days during the preceding 30 days.

BRFSS asks respondents how many days during the preceding 30 days their mental health was not good, including stress, depression, and emotional problems.

Decisions & Implications

Crosssectional survey data	Each BRFSS year surveys **different respondents.** The models identify patterns associated with reported mental distress, but cannot predict future outcomes for specific individuals.
Population definitions	The primary comparison focuses on AI/AN-alone adults and other identified single-race adults. **Multiracial and unresolved classifications were excluded** from this comparison because AI/AN combinations **could not be consistently identified** across years. This limits population representation.
Historical data harmonization	Survey questions, variable names, and response codes changed between 2014 and 2025. **Variables were standardize**d where possible, and information that could not be compared consistently was excluded or simplified.
Predictor selection	Six consistently available predictors were selected: **physical-health days, general health, age, sex, education, and income**. Some self reported behaviors were excluded due to sensitivity with social desirability / recall bias. This focuses the model but also limits the range of factors the models can evaluate.
Survey weighting	CDC BRFSS **survey weights** (_LLCPWT) and calculated weights were used in population estimates and model evaluations to account for unequal selection probabilities and support more representative estimates.

2025: missing predictor categories retained in modeling



Missing Predictor Data

Household income had the greatest missingness (~19%) in both populations..

Physically unhealthy days had higher missingsness among AI/AN respondents ~3.4% vs. ~2.3%).

Differences in data completeness, observed mental Distress prevalence reinforce the importance of evaluating model reliability separately across populations.

Research Hypotheses

H1 | Predictive Performance

Hypothesis:
Machine learning models trained on earlier BRFSS survey data will make more accurate predictions of frequent mental distress than a model based only on historical prevalence.

Null hypothesis (H1o):
The machine learning model will not perform better than the historical baseline.

Alternative hypothesis (H1a):
The machine learning model will perform better than the historical baseline.

Evaluation Metric:
Survey-weighted Brier score (MSE). Lower scores indicate better accuracy.

H2 | Population-Specific Model Reliability

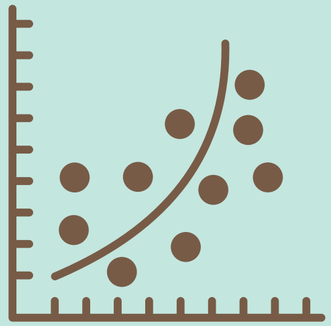
Hypothesis:
The nationally trained model will differ in its ability to distinguish frequent mental distress between the two populations.

Null hypothesis (H2o):
The model's ability to predict FMD is the same across both populations.

Alternative hypothesis (H2a):
The model's ability to predict FMD differs between populations.

Evaluation Metric:
Survey-weighted AUROC, difference between populations evaluated at 95% CI.

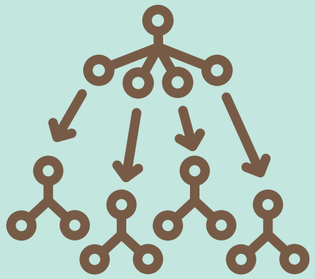
Learning Models



Logistic Regression
Interpretable statistical classifier



LASSO
Regularized logistic regression



Random Forest
Nonlinear ensemble classifier

Prediction Target
Frequent Mental Distress



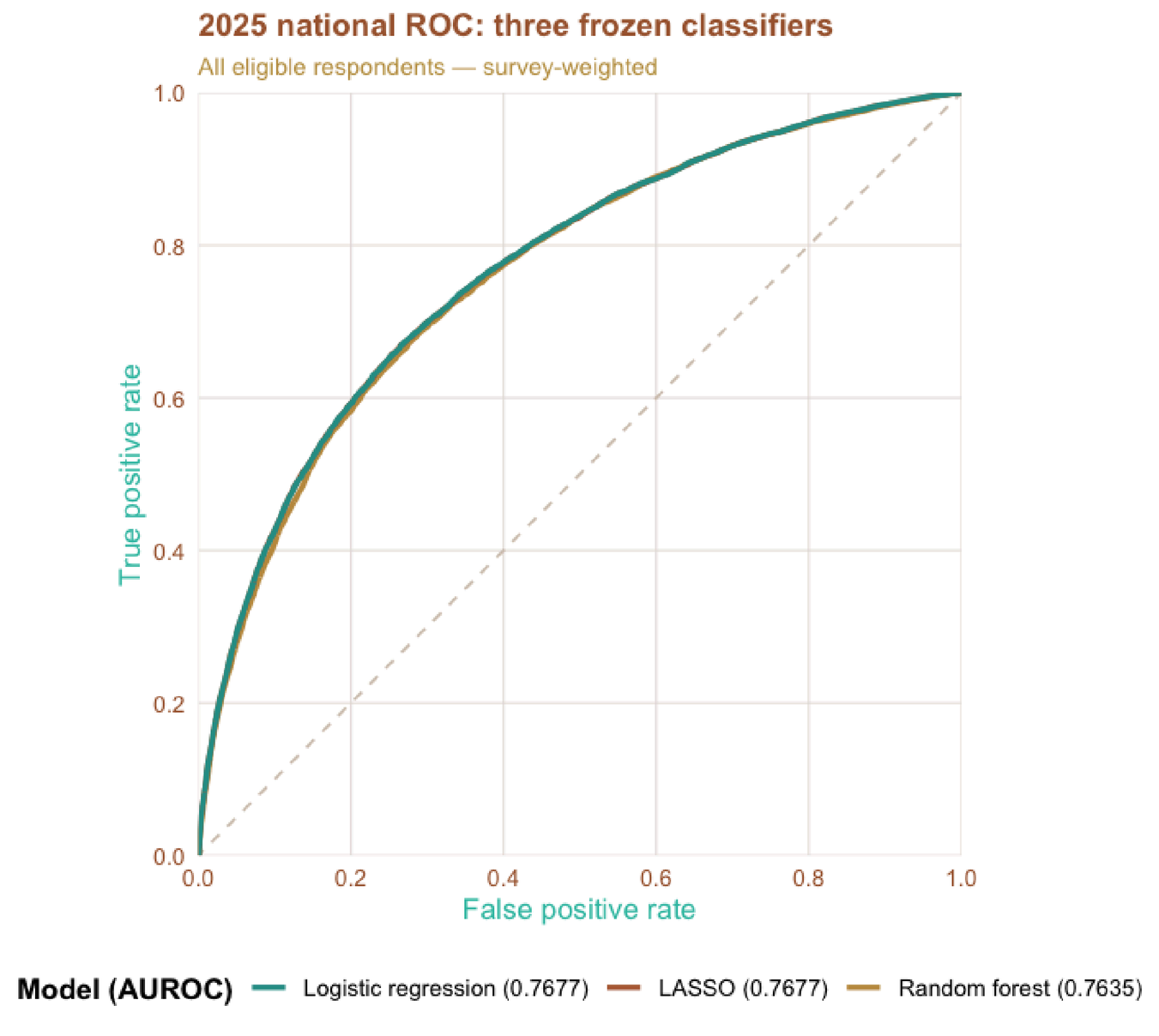
Data Split | Survey Years

2014 - 2022
Development
Train and validate models

2023 Selection
Compare model families

2024 Calibration
Adjust probabilities

2025 Evaluation
Evaluate prediction performance



At a 0.50 threshold, sensitivity was 17.4%; the model is better suited to evaluating relative probabilities than identifying all FMD cases at this cutoff.

Machine Learning Outperforms Historical Baseline

349,674 respondents with known FMD outcomes

All three models demonstrated meaningful discrimination, with logistic regression and LASSO performing almost identically. Predicted probabilities improved on the historical-prevalence benchmark.

Logistic Regression Evaluation Metrics

The selected model can distinguish FMD from non-FMD better than chance (0.50).

.7677

Survey-weighted AUROC

Predictions have lower squared probability error than the historical baseline (0.13656).

.11447

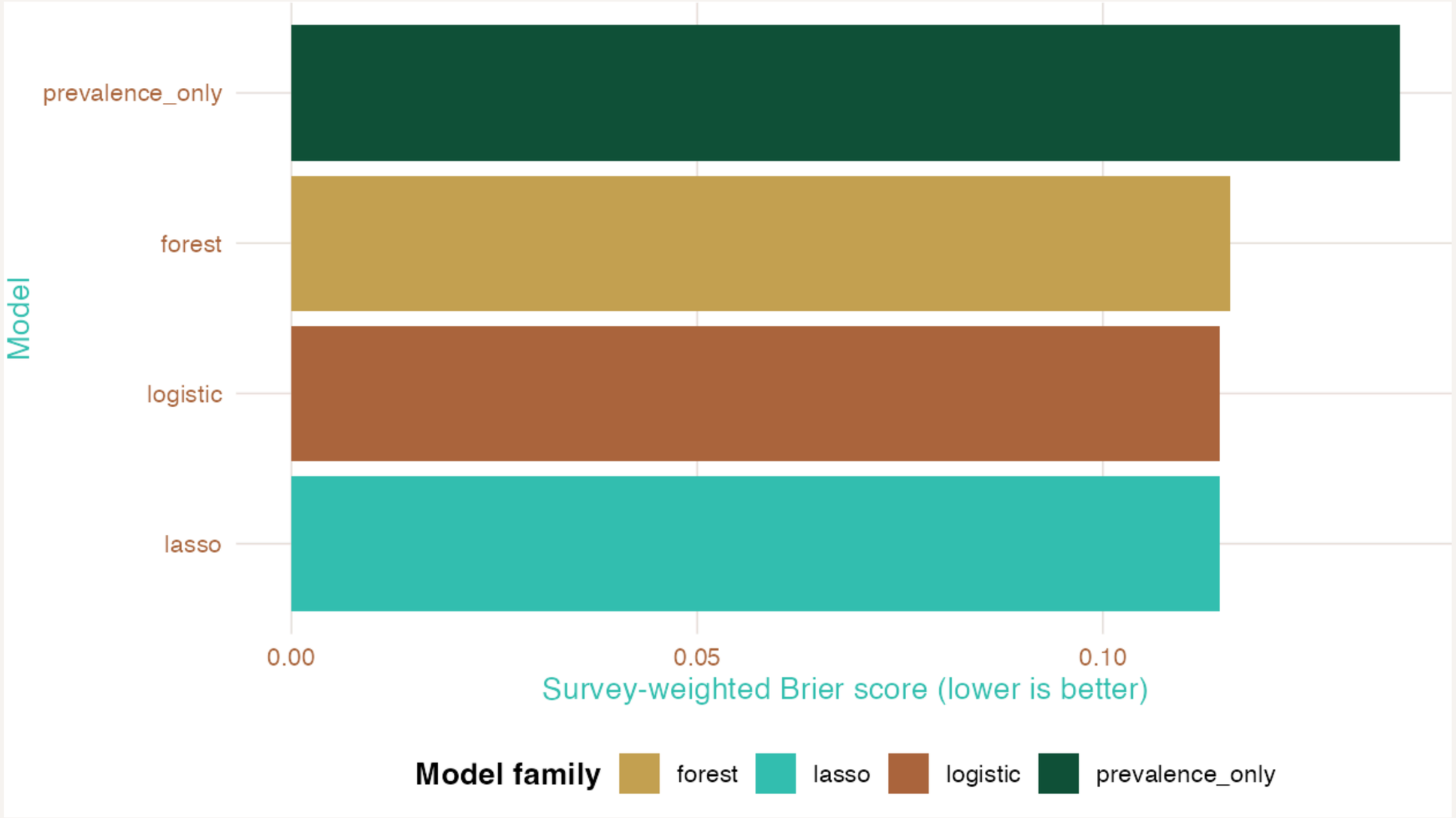
Brier score

The model ranks FMD cases substantially better than the prevalence-based reference of approximately 0.1617.

0.4255

Average Precision

General Model Performance on 2025 Data



LASSO had the lowest 2025 Brier score by just 0.00001; logistic regression was selected on 2023 data.

question 1

Can we predict frequent mental distress?

Can models trained on earlier BRFSS data accurately classify self-reported frequent mental distress among respondents in later survey years?

We reject the Null Hypothesis

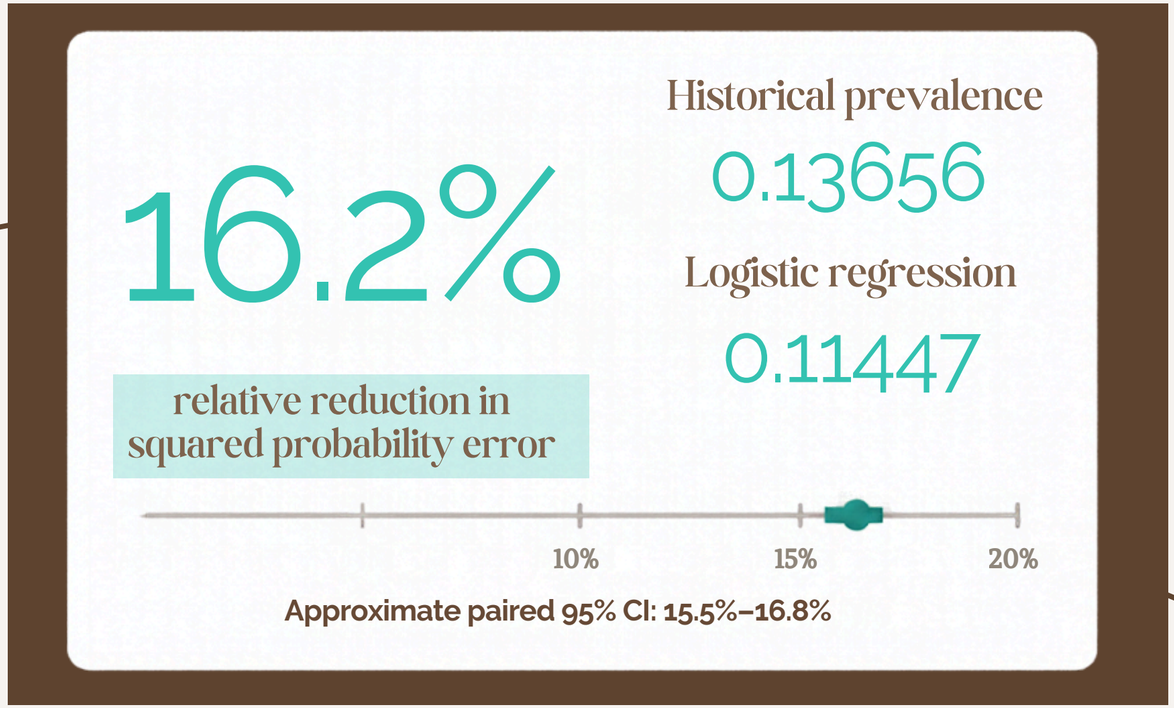
H1 | Predictive Performance

Hypothesis:
Machine learning models trained on earlier BRFSS survey data will make more accurate predictions of frequent mental distress than a model based only on historical prevalence.

Null hypothesis (H0):
The machine learning model will not perform better than the historical baseline.

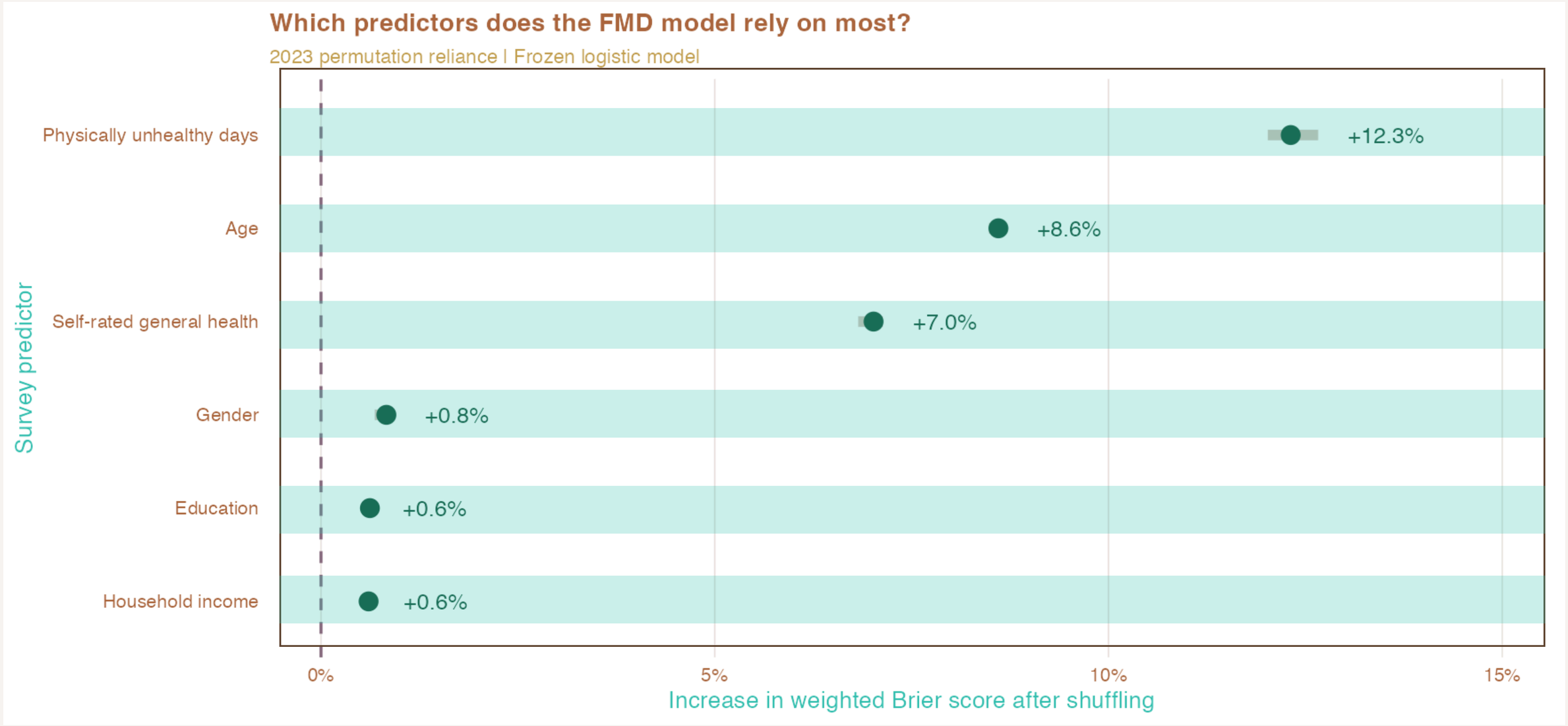
Alternative hypothesis (Ha):
The machine learning model will perform better than the historical baseline.

Evaluation Metric:
Survey-weighted Brier score (MSE). Lower scores indicate better accuracy.



The selected logistic regression model reduced squared probability error by **16.2%** compared with the historical-prevalence baseline.

The approximate paired **95% confidence interval (15.5%–16.8%)** falls entirely above zero, providing evidence of improved predictive accuracy at the **5% significance level**.


Which predictors does the FMD model rely on most?
2023 permutation reliance | Frozen logistic model
Survey predictor
Physically unhealthy days
Age
Self-rated general health
Gender
Education
Household income
+12.3%
+8.6%
+7.0%
+0.8%
+0.6%
+0.6%
0%
5%
10%
15%
Increase in weighted Brier score after shuffling

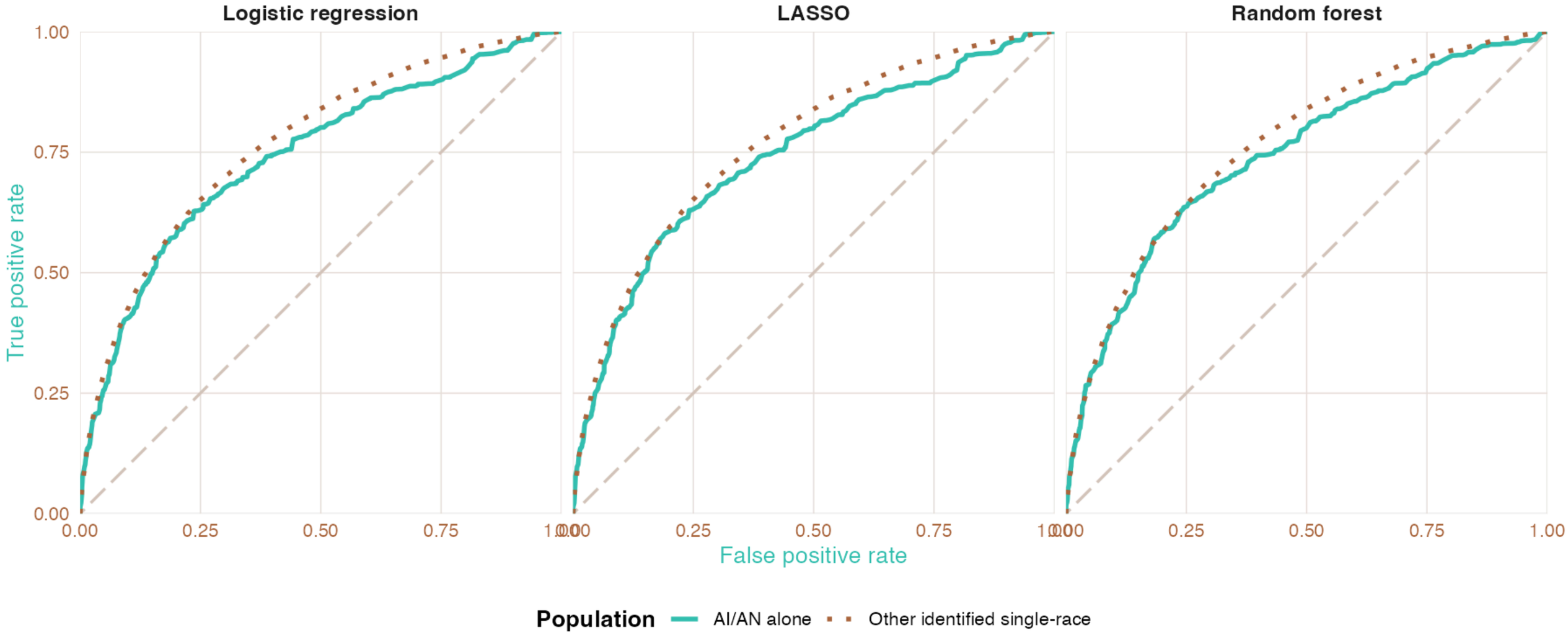
question 2

How well does the model work for AI/AN adults?

How does the same nationally trained model perform for AI/AN- adults (p1) compared with other self-identified single-race adults (p2)?

2025 model discrimination by population

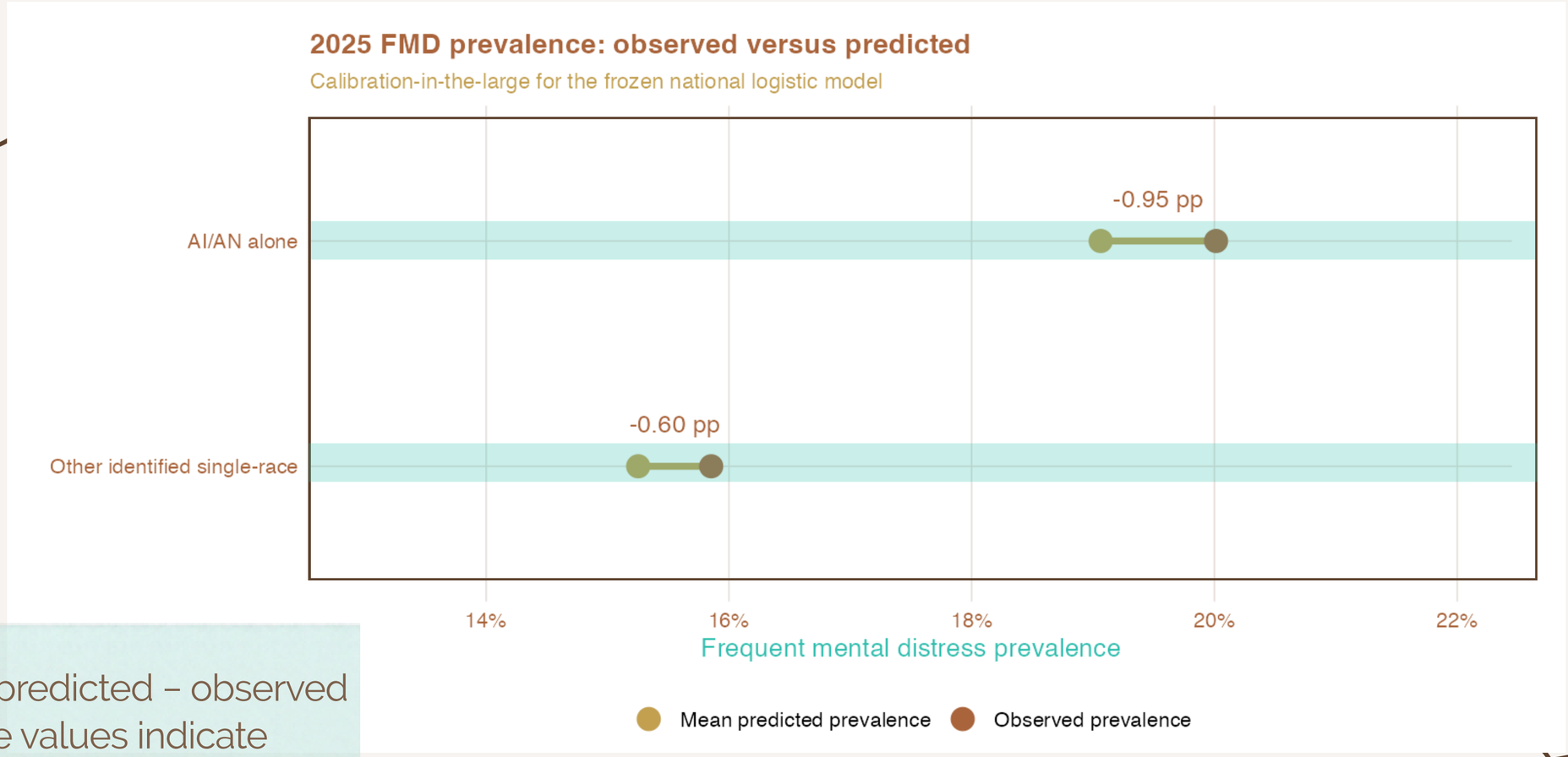
Survey-weighted ROC curves for three historically trained classifiers



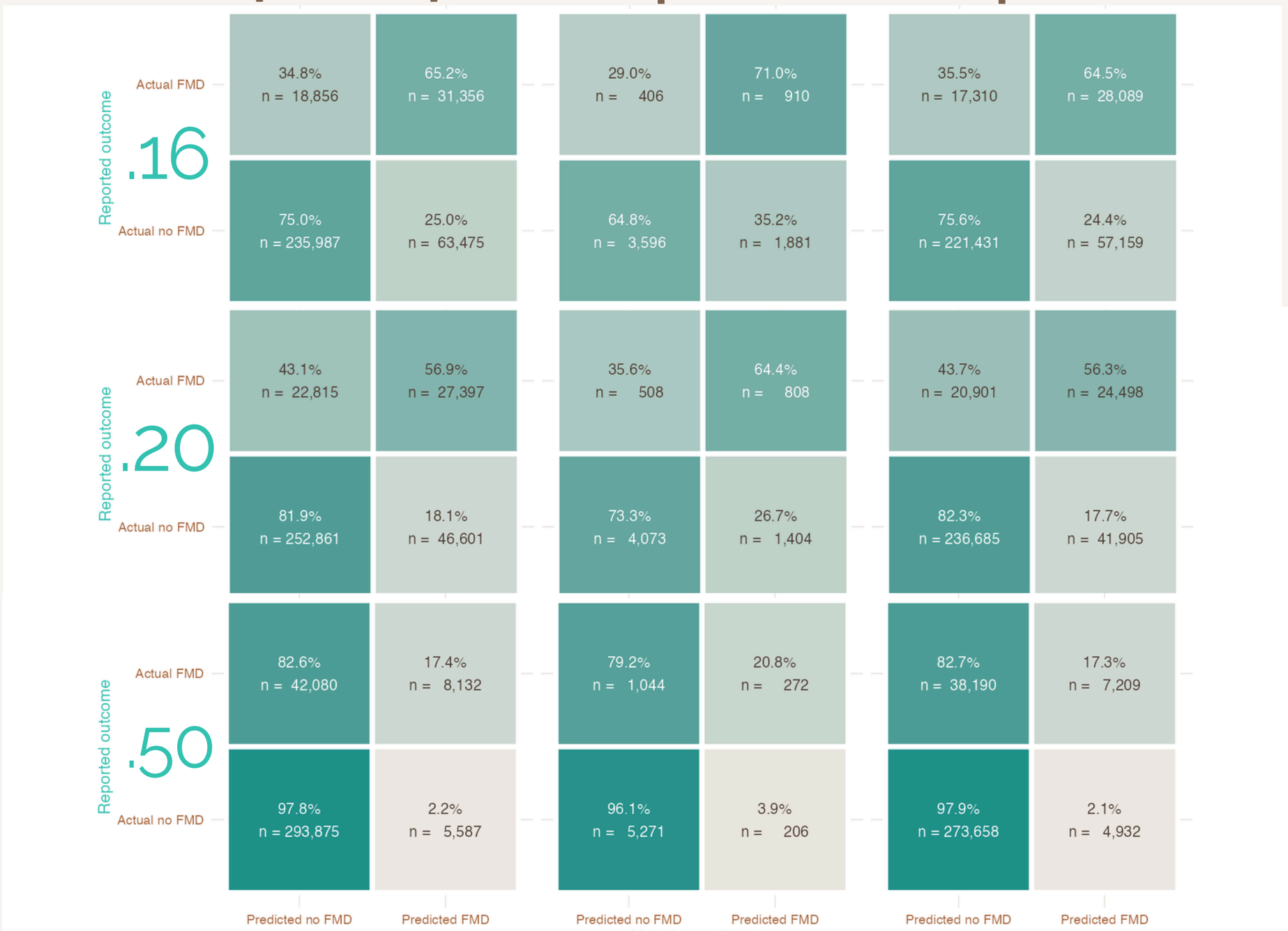
How Well Does the Model Distinguish FMD Across Groups?

AI/AN AUROC 0.7413

Comparison AUROC 0.7684



Difference = predicted − observed
negative values indicate
underestimation



F1 scores at .16

p1 and p2	.432
p1	.443
p2	.430

F1 scores at .20

p1 and p2	.441
p1	.458
p2	.438

F1 scores at .50

p1 and p2	.254
p1	.303
p2	.251

question 2

How well does the model work for AI/AN adults?

How does the same nationally trained model perform for AI/AN-adults compared with other self-identified single-race adults?

Inconclusive

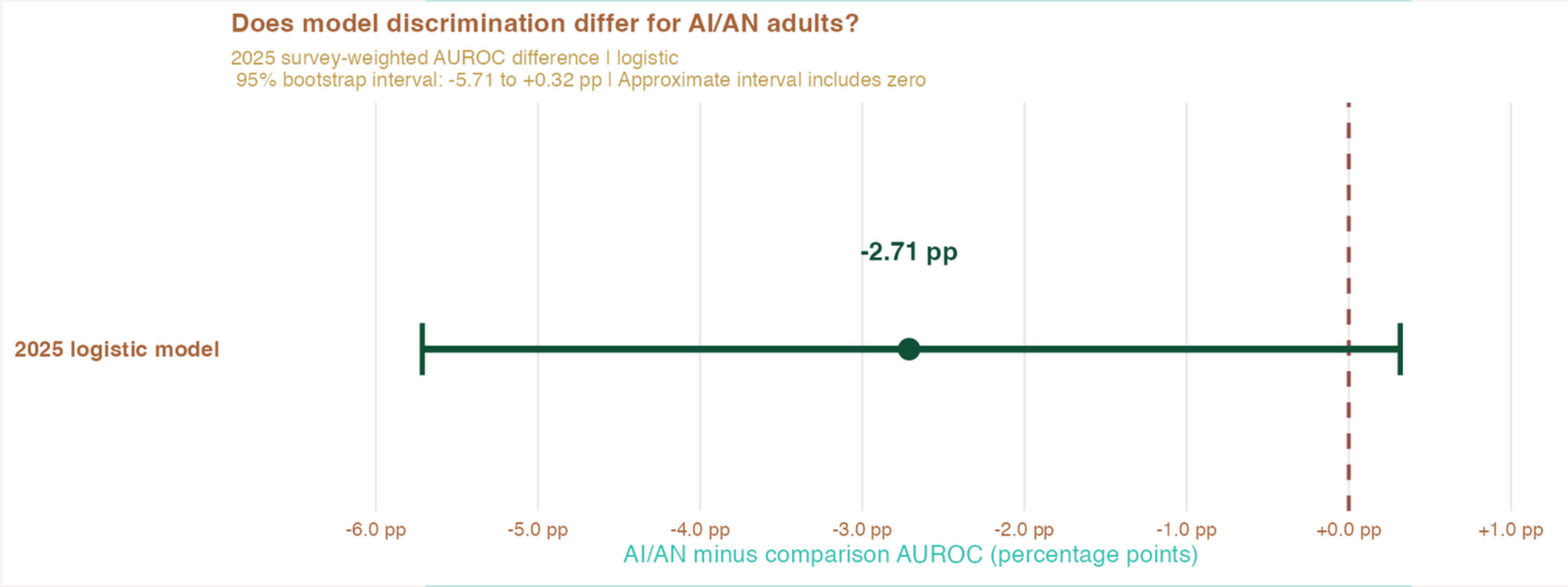
H2 | Population-Specific Model Reliability

Hypothesis:
The nationally trained model will differ in its ability to distinguish frequent mental distress between the two populations.

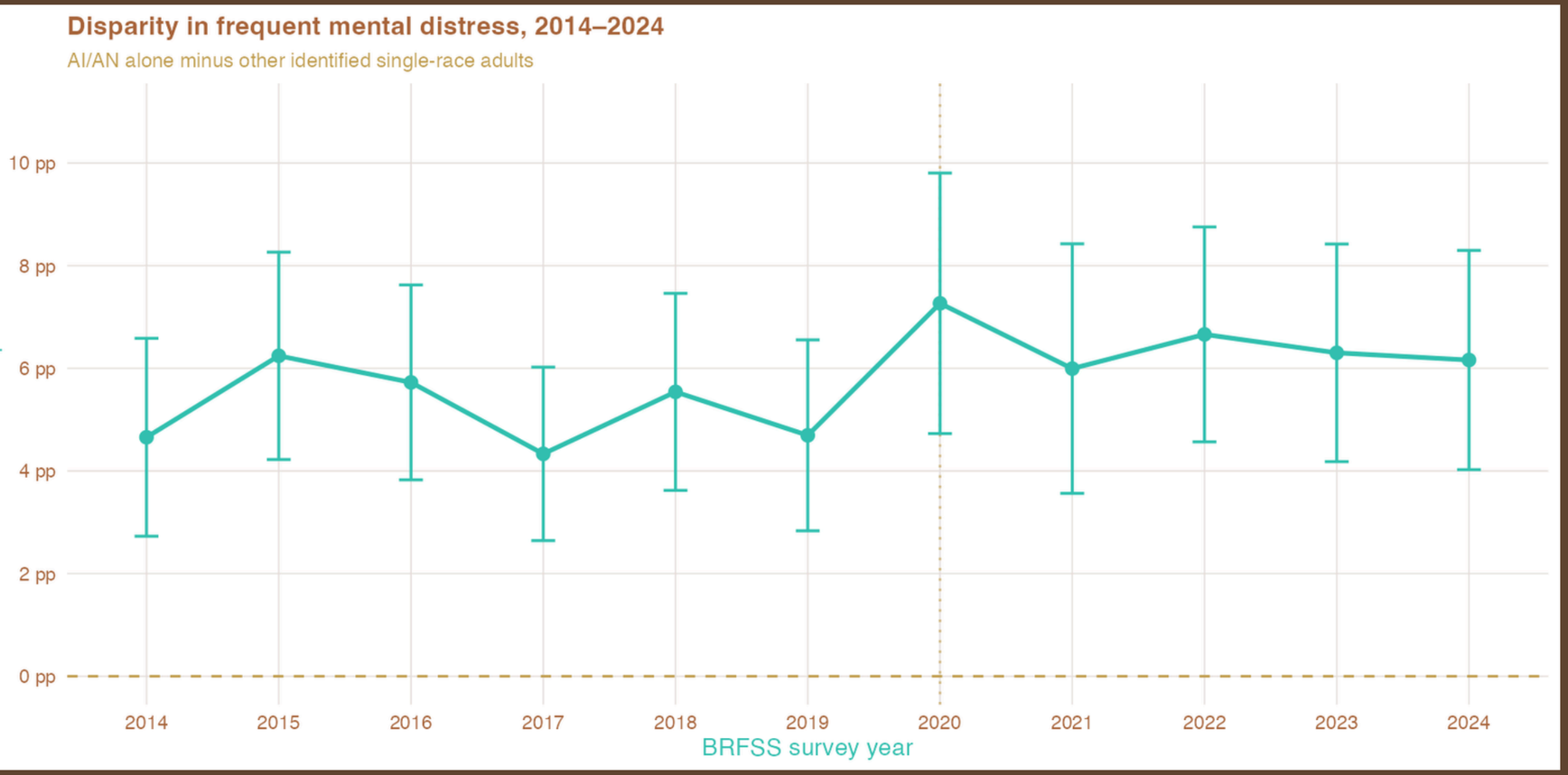
Null hypothesis (HO):
The model's ability to predict FMD is the same across both populations.

Alternative hypothesis (Ha):
The model's ability to predict FMD differs between populations.

Evaluation Metric:
Survey-weighted AUROC, difference between populations evaluated at 95% CI.



AUROC was estimated to be 2.71 percentage points lower for p1 than for p2, but the 95% confidence interval included zero, leaving the difference inconclusive.



People across populations may share many of the same factors associated with mental distress. But shared predictors do not necessarily mean equal prevalence: **one population may experience those factors more often** or face additional circumstances that the model does not capture. Our results do not establish why prevalence differs.

What Did We Unravel?



H1 | Supported	• **16.2%** lower national Brier error than historical prevalence alone • Approximate **95% CI: 15.5%–16.8%**
H2 | Inconclusive	• Prediction discrimination for p1 was lower in the observed results than for p2 • The confidence **interval included zero,** so unequal performance was not established.
Key predictor insight	• Physical health emerged as a major predictor • Physically unhealthy days, age, and general health were important to model performance



Miigwitch!

Bogic, M., Hebert, L. E., Evanson, A., Wright, B., Fruhbauerova, M., Petras, A., Jansen, K., Shaw, J., Bradshaw, S., O'Leary, M., Zacher, T., Smoker, K., Comtois, K. A., & Nelson, L. (2024). Connected for life: How social connectedness can help prevent suicide in American Indian and Alaska Native communities. Archives of Psychiatric Nursing, 51, 259–267. https://doi.org/10.1016/j.apnu.2024.06.006

Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE principles for Indigenous data governance. Data Science Journal, 19, Article 43. https://doi.org/10.5334/dsj-2020-043

Centers for Disease Control and Prevention. (2024a, October 29). Health status. PLACES. https://www.cdc.gov/places/measure-definitions/health-status.html

Centers for Disease Control and Prevention. (2024b). Mental health. Chronic Disease Indicators. https://www.cdc.gov/cdi/indicator-definitions/mental-health.html

Centers for Disease Control and Prevention. (2025a). 2024 BRFSS survey data and documentation [Data set]. https://www.cdc.gov/brfss/annual_data/annual_2024.html

Centers for Disease Control and Prevention. (2025b). Complex sampling weights and preparing 2024 BRFSS module data for analysis. https://www.cdc.gov/brfss/annual_data/2024/pdf/Complex-Sampling-Weights-and-Preparing-Module-Data-for-Analysis-2024-508.pdf

Centers for Disease Control and Prevention. (2026a). 2025 BRFSS survey data and documentation [Data set]. https://www.cdc.gov/brfss/annual_data/annual_2025.html

Centers for Disease Control and Prevention. (2026b, September 8). Annual survey data. Behavioral Risk Factor Surveillance System. https://www.cdc.gov/brfss/annual_data/annual_data.htm/

Centers for Disease Control and Prevention. (2026c). Suicide data and statistics. https://www.cdc.gov/suicide/data/

Centers for Disease Control and Prevention. (2026d). Tribal suicide prevention. https://www.cdc.gov/suicide/program/tribal-suicide-prevention.html

Federal Reserve Bank of Minneapolis, Center for Indian Country Development. (n.d.). About the data. https://www.minneapolisfed.org/indiancountry/resources/native-economic-trends/about-the-data

Fetter, A. K., Wiglesworth, A., Rey, L. F., Azarani, M., Prairie Chicken, M. L., Young, A. R., Riegelman, A., & Gone, J. P. (2023). Risk factors for suicidal behaviors in American Indian and Alaska Native peoples: A systematic review. Clinical Psychological Science, 11(3), 528–551. https://doi.org/10.1177/21677026221126732

Gartner, D. R., & Martinez, R. A. M. (2025). Reflecting on power: The inadequacies of small sample size technique through the lens of Indigenous health. American Journal of Epidemiology, 194(9), 2524–2531. https://doi.org/10.1093/aje/kwaf097

Indian Health Service. (n.d.-a). About. Suicide Prevention and Care Program. https://www.ihs.gov/suicideprevention/about/

Indian Health Service. (n.d.-b). Chapter 34: Suicide prevention and care. Indian Health Manual, Part 3. https://www.ihs.gov/ihm/pc/part-3/p3c34/

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. Science, 366(6464), 447–453. https://doi.org/10.1126/science.aax2342

Office of Disease Prevention and Health Promotion. (n.d.). Social determinants of health. Healthy People 2030. U.S. Department of Health and Human Services. https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health

Office of Minority Health. (n.d.). Mental health and American Indians/Alaska Natives. U.S. Department of Health and Human Services. Retrieved November 8, 2026, from https://minorityhealth.hhs.gov/mental-and-behavioral-health-american-indiansalaska-natives

Pattee, J. (2026). Guided project [Course materials and programming code]. AIMAHEAD AIHEC TCU Student Readiness Program.

Singh, G. K., Lee, H., Kim, L. H., & Williams, S. D. (2024). Social determinants of health among American Indians and Alaska Natives and tribal communities: Comparison with other major racial and ethnic groups in the United States, 1990–2022. International Journal of Maternal and Child Health and AIDS, 13, Article e010. https://doi.org/10.25259/IJMA_10_2024

Stone, D., Trinh, E., Zhou, H., Welder, L., End of Horn, P., Fowler, K., & Ivey-Stephenson, A. (2022). Suicides among American Indian or Alaska Native persons—National Violent Death Reporting System, United States, 2015–2020. Morbidity and Mortality Weekly Report, 71(37), 1161–1168. https://doi.org/10.15585/mmwr.mm7137a1

Substance Abuse and Mental Health Services Administration. (2025). Key substance use and mental health indicators in the United States: Results from the 2024 National Survey on Drug Use and Health (HHS Publication No. PEP25-07-007, NSDUH Series H-60). Center for Behavioral Health Statistics and Quality. https://www.samhsa.gov/data/sites/default/files/reports/rpt56287/2024-nsduh-annual-national-report.pdf